

PriceFM: Foundation Model for Probabilistic Electricity Price Forecasting

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Abstract

Electricity price forecasting in Europe presents unique challenges due to the continent’s increasingly integrated and physically interconnected power market. While recent advances in foundation models have led to substantial improvements in general time series forecasting, most existing approaches do not incorporate prior graph knowledge from the transmission topology, which can limit their ability to exploit meaningful cross-region dependencies in interconnected power systems, motivating a domain-specific foundation model. In this paper, we address this gap by first introducing a comprehensive and up-to-date dataset across 24 European countries (38 regions), spanning from 2022-01-01 to 2026-01-01. Building on this groundwork, we propose PRICEFM, a probabilistic foundation model pretrained on this large dataset. Specifically, PRICEFM maps each region’s price and exogenous features into a comparable latent embedding via a shared Mixture-of-Experts (MoE) projection layer, then injects prior graph knowledge by constructing a sparse graph mask derived from transmission topology. Across a large-scale European benchmark, PriceFM achieves strong performance and demonstrates superior generalization under both zero-shot and full-shot evaluation compared with multiple competitive baselines.

1. Introduction

The European electricity market is physically interconnected through a network of cross-border transmission lines, enabling the exchange of electricity between regions (Lago et al., 2018). However, physical constraints, such as limited transmission capacity, can restrict electricity flow between regions and lead to zonal price differences (Finck, 2021),

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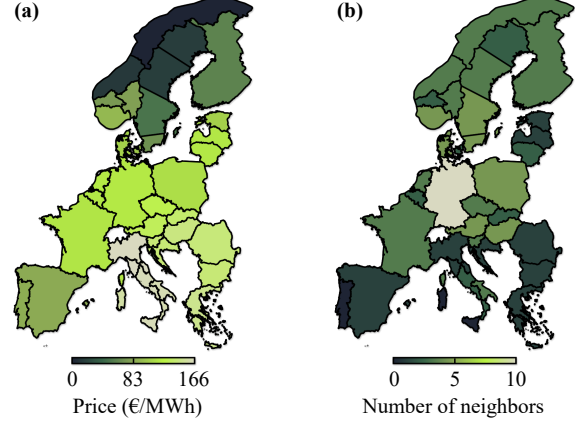


Figure 1. Spatial distribution of electricity price and number of neighboring regions. (a) Electricity prices for 38 European regions averaged from 2022-01-01 to 2026-01-01. A significant zonal price difference is observed between north and south regions. (b) Number of neighboring regions that are *directly* connected to certain region via transmission lines. For example, France and Portugal are connected to Spain, thus the number of neighboring regions for Spain is 2. The mean value across all regions is 3.4.

illustrated in Figure 1. These price disparities highlight the spatial nature of electricity price formation. Recent studies show that electricity price dynamics are strongly influenced by spatial interdependencies and cannot be accurately captured using region-specific models (Do et al., 2024; Yu et al., 2026a). Therefore, explicitly modeling the spatial structure of the European electricity market is essential for producing accurate price forecasts.

Most existing studies on electricity price forecasting do not explicitly model the spatial structure and focus on a single-region market, particularly Germany (Munian & Ziel, 2020; Maciejowska et al., 2021; Kitsatoglou et al., 2024), as the German market is one of the largest markets in Europe. Other studies explore forecasting methods for markets such as Denmark, Finland, Spain, and Austria, also using region-specific models (Ziel & Weron, 2018; Gianfreda et al., 2020; Loizidis et al., 2024; Yu et al., 2026c). More recent works explicitly model the spatial nature of the electricity price.

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For instance, a Graph Convolutional Network (GCN) is applied to capture spatial interdependencies in the Nordic markets, such as Norway, Sweden, and Finland (Yang et al., 2024). Moreover, an attention-based variant is developed to predict prices in certain European markets such as Austria, Germany, and Hungary (Meng et al., 2024). However, these models cover only subsets of Europe and learn spatial dependencies through fully learnable mechanisms (e.g., spatial convolutions or self-attention). Such designs may inadvertently incorporate signals from topologically distant regions that are weakly related to the target region, introducing noise and increasing the overfitting risk. This motivates incorporating transmission-topology graph knowledge as an explicit regularization to constrain spatial information flow and improve generalization.

Unlike conventional forecasting models trained from scratch, time-series foundation models have achieved remarkable success across diverse domains such as weather, transportation, and energy, by capturing complex temporal patterns and exhibiting strong generalization capabilities (Ansari et al., 2024; Das et al., 2024; Liu et al., 2025; Shi et al., 2025). However, electricity prices are shaped not only by local fundamentals but also by signals from neighboring regions through the transmission lines. Existing foundation models combine time series through purely data-driven mixing without incorporating transmission-topology priors, and therefore cannot exploit physically meaningful spatial structure. This gap motivates a domain-specific foundation model with injected graph prior.

To support the development of a domain-specific foundation model for electricity price forecasting, there is a pressing need for high-quality, large-scale, and up-to-date datasets that reflect the spatiotemporal complexity of integrated European markets. However, existing datasets are often fragmented in structure, cover only short time periods, are outdated, or focus on individual regions (Lago et al., 2021). This lack of standardized data poses a significant barrier to training and evaluating domain-specific foundation models. We address these limitations by introducing a comprehensive and up-to-date dataset and proposing PriceFM, a foundation model that utilizes a shared MoE projection layer to process multi-region inputs and regularizes noisy signals from distant regions via a topology-guided sparse graph mask. In summary, our contributions are as follows:

- We introduce a comprehensive and up-to-date dataset. To the best of our knowledge, this is the largest and most diverse open dataset for European electricity markets, comprising day-ahead electricity prices, day-ahead forecasts of load, solar, and wind power generation, covering 24 European countries (38 regions), spanning from 2022-01-01 to 2026-01-01.
- We propose and release the PriceFM, a novel fore-

casting framework that integrates prior graph knowledge derived from the transmission topology of the European electricity market. PriceFM supports multi-region, multi-timestep, and multi-quantile forecasting.

- We conduct experiments to evaluate the model’s performance against multiple baselines, and assess the impact of design choices through ablation studies, thereby providing both quantitative evidence of overall performance and insights into optimal configurations.

2. Related Work

Foundation Models. Foundation models are typically pretrained on large-scale datasets and then transferred to new tasks in a zero-shot manner. Representative examples include Chronos (Ansari et al., 2024), TimesFM (Das et al., 2024), Moirai (Liu et al., 2025), and TimeMoE (Shi et al., 2025). Pretraining enables these models to learn reusable temporal representations and to generalize across domains without retraining on the target dataset. This property makes them suitable baselines for evaluating *zero-shot* price forecasting, where the target regions are not used for training.

Time-Series Models. Generic time-series models can be trained from scratch and applied across a wide range of time-series tasks. Representative examples include FEDFormer (Zhou et al., 2022), iTransformer (Liu et al., 2023), PatchTST (Nie et al., 2023), TimesNet (Wu et al., 2023), and TimeXer (Wang et al., 2024). Although these methods are not necessarily pretrained as foundation models, they often achieve strong performance when trained end-to-end on the target dataset, serving as competitive baselines for *full-shot* evaluation in electricity price forecasting.

Graph Models. Graph-based models explicitly represent spatial structure by modeling regions as nodes and their relations as edges, enabling information propagation across the graph. Representative examples include Graph Convolutional Network (GCN) (Kipf, 2016), Graph Attention Network (GAT) (Veličković et al., 2017), GraphSAGE (Hamilton et al., 2017), GraphDiffusion (Li et al., 2018), and GraphARMA (Bianchi et al., 2021). By incorporating an adjacency matrix, these models can learn spatial mixing patterns jointly with temporal dynamics. This property makes them suitable baselines for evaluating whether injecting a topology-constrained sparse graph prior improves multi-region electricity price forecasting, compared with purely learned spatial representations under the *full-shot* setting.

3. Preliminary

The forecasting target is a probabilistic price trajectory, i.e., $\mathcal{T} = 96$ quarter-hourly prices for the deliv-

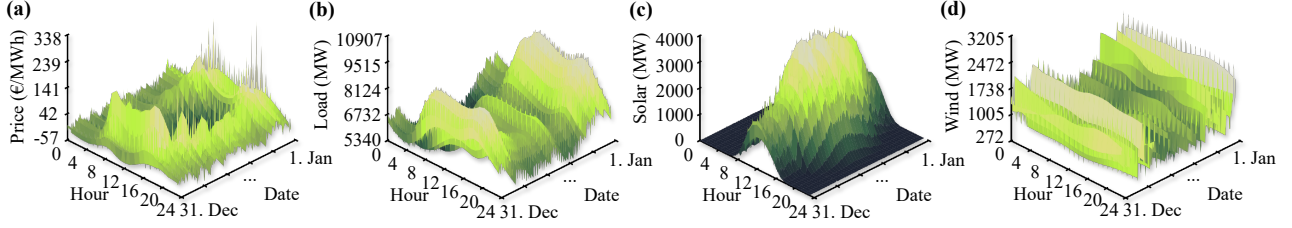


Figure 2. European-level energy data in 2025, averaged across regions. (a) Electricity price. Price spikes sharply during the morning and evening peak, dip around midday, and shows higher volatility in the winter. (b) Forecasted load. Load exhibits a double-peak each day with winter peaks substantially larger than summer. (c) Forecasted solar power generation. Solar is zero overnight, rises in a smooth bell curve to a strong midday maximum, then falls back to zero by dusk, and is much higher in summer. (d) Forecasted wind power generation (onshore and offshore). Wind lacks a daily pattern, fluctuates with high-frequency spikes, and is much higher in winter.

ery day $\mathcal{D} + 1$ with a set of quantiles ($\tau \in \mathcal{Q} = \{0.10, 0.25, 0.45, 0.50, 0.55, 0.75, 0.90\}$), using data available before gate closure, typically around midday on day $\mathcal{D}$. After midday on $\mathcal{D}$, the electricity prices for $\mathcal{D} + 1$ are published and known. We employ a backward-looking window of size L (e.g. $L = 96$ corresponds to 24 hours from $\mathcal{D}$), for known electricity prices, denoted as $\mathbf{X}_{r_{in}}^{\text{price}}$. We also include forward-looking exogenous features, such as day-ahead forecasts of load, solar, and wind (sum of onshore and offshore) power generation for $\mathcal{D} + 1$, denoted as $\mathbf{X}_{r_{in}}^{\text{exo}}$, made on $\mathcal{D}$ before gate closure, as well as their historical values over L . The forecasting setup and the choice of feature set are widely used in prior works (Maciejowska, 2020; Uniejewski & Weron, 2021; Meng et al., 2024). Importantly, this work utilizes multi-region inputs to produce multi-region, multi-timestep, and multi-quantile forecasts. Therefore, the input and output of PRICEFM are defined as:

- **Input:** $\{\mathbf{X}_{r_{in}}^{\text{price}}, \mathbf{X}_{r_{in}}^{\text{exo}}\}_{r_{in} \in \mathcal{R}}$,
where $\mathbf{X}_{r_{in}}^{\text{price}} \in \mathbb{R}^{L \times f_1}$ and $\mathbf{X}_{r_{in}}^{\text{exo}} \in \mathbb{R}^{(L+\mathcal{T}) \times f_2}$,
- **Output:** $\{\hat{\mathbf{y}}_{r_{out}, \tau}\}_{r_{out} \in \mathcal{R}, \tau \in \mathcal{Q}}$,
where $\hat{\mathbf{y}}_{r_{out}, \tau} \in \mathbb{R}^{\mathcal{T}}$,

where $r_{in}, r_{out} \in \mathcal{R} = \{\text{AT}, \dots, \text{SK}\}$ are region codes (detailed in Appendix, Table 4), $f_1 = 1$, and $f_2 = 3$.

4. Data

4.1. Spatiotemporal Coverage

Spatially, the dataset covers 24 European countries (38 regions). These regions reflect transmission zones rather than administrative boundaries. For example, Denmark (DK) is split into two regions: DK1 and DK2. Each is connected to different regions, resulting in distinct cross-border power flows. Temporally, the dataset spans from 2022-01-01 to 2026-01-01, providing wide temporal coverage.

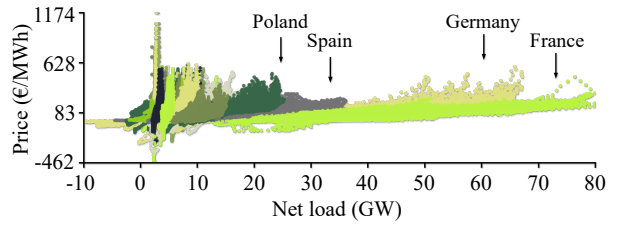


Figure 3. Price–net load relationship across European regions (Net load = load − solar − wind). Distinct regional behaviors are evident in markets such as Germany, France, Spain, and Poland. In contrast, other regions share similar patterns.

4.2. Feature Set

The feature set includes day-ahead electricity prices, load forecasts, and solar and wind power generation forecasts, where the wind feature is computed by summing the offshore and onshore wind power generation. For simplicity, we refer to these features as *price*, *load*, *solar*, and *wind*, respectively. The availability of features across regions is detailed in Appendix, Table 4. A European-level visualization of these features is shown in Figure 2.

4.3. Resolution

We resample all features to a 15-min resolution for two reasons: (1) an increasing number of EU electricity markets are moving from 60-min resolution to 15-min resolution; and (2) the raw data exhibit heterogeneous temporal resolutions. For example, load in Spain is provided hourly before 2022-05-23 and then switches to quarter-hourly resolution afterward; in Austria, load is reported quarter-hourly while prices are with hourly resolution before 2025-10-01.

4.4. Missing Value

Partial features are excluded due to the high rate of missing values ($> 20\%$), summarized in Appendix, Table 4. For

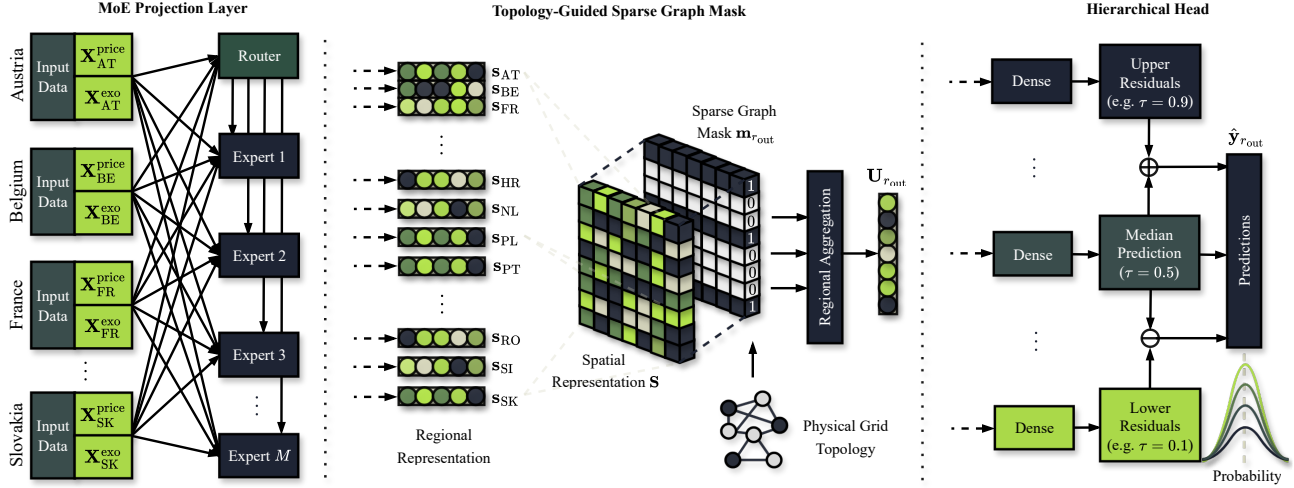


Figure 4. Structure of PriceFM. The input features $\mathbf{X}_{r_{in}}^{\text{price}}$ and $\mathbf{X}_{r_{in}}^{\text{exo}}$ are passed into a MoE projection layer to produce the regional representations. The regional representations are stacked to form the shared spatial representation $\mathbf{S}$, which is multiplied with the sparse graph mask to produce the spatial representation $\mathbf{U}_{r_{out}}$. $\mathbf{U}_{r_{out}}$ is fed into hierarchical quantile heads to produce probabilistic forecasts.

example, solar from Latvia has a 56.6% missing rate and is only available after 2024-04-07. The features with low missing rates ($< 1\%$) are filled using linear interpolation. If a region does not provide a certain generation type (e.g., wind), we keep the input dimensionality fixed by adding an all-zero feature, indicating that no such generation is produced in that region.

5. PriceFM

5.1. MoE Projection Layer

As later introduced in Section 5.2, we will inject graph knowledge to compute price representations across regions. This requires that the regional price representations are *comparable* and lie in a shared embedding space.

A natural solution is to assign 38 dense layers to 38 input regions. However, as shown in Fig. 3, some regions exhibit similar patterns, suggesting that they can share parts of the projection mechanism. To this end, we design a *shared* Mixture-of-Experts (MoE) projection layer that maps each region's inputs ($\mathbf{X}_{r_{in}}^{\text{price}}, \mathbf{X}_{r_{in}}^{\text{exo}}$) into a regional representation.

Fusion Expert. We reshape each modality into a latent embedding of dimension h via a dense layer with *Swish* activation, and inject the exogenous representation as a residual into the price representation:

$$\mathbf{X}_{r_{in}}^{\text{price}} \xrightarrow{\text{Project}} \hat{\mathbf{z}}_{r_{in}}^{\text{price}} \in \mathbb{R}^h, \quad (1)$$

$$\mathbf{X}_{r_{in}}^{\text{exo}} \xrightarrow{\text{Project}} \hat{\mathbf{z}}_{r_{in}}^{\text{exo}} \in \mathbb{R}^h, \quad (2)$$

$$\mathbf{z}_{r_{in}} = \text{Swish}(\hat{\mathbf{z}}_{r_{in}}^{\text{price}} + \hat{\mathbf{z}}_{r_{in}}^{\text{exo}}) \in \mathbb{R}^h. \quad (3)$$

Weighting Router. Similar to the fusion expert in Eq. (3), the router takes the same pair of inputs ($\mathbf{X}_{r_{in}}^{\text{price}}, \mathbf{X}_{r_{in}}^{\text{exo}}$), but uses a dense layer with *softmax* activation to output the expert weights:

$$\mathbf{X}_{r_{in}}^{\text{price}} \xrightarrow{\text{Project}} \hat{\pi}_{r_{in}}^{\text{price}} \in \mathbb{R}^M, \quad (4)$$

$$\mathbf{X}_{r_{in}}^{\text{exo}} \xrightarrow{\text{Project}} \hat{\pi}_{r_{in}}^{\text{exo}} \in \mathbb{R}^M, \quad (5)$$

$$\pi_{r_{in}} = \text{Softmax}(\hat{\pi}_{r_{in}}^{\text{price}} + \hat{\pi}_{r_{in}}^{\text{exo}}) \in \mathbb{R}^M. \quad (6)$$

Let M denote the number of experts. Given $\mathbf{z}_{r_{in}} \in \mathbb{R}^h$, the M experts output an *expert matrix*:

$$\mathbf{Z}_{r_{in}} = \begin{bmatrix} (\mathbf{z}_{r_{in}})_1 \\ (\mathbf{z}_{r_{in}})_2 \\ \vdots \\ (\mathbf{z}_{r_{in}})_M \end{bmatrix} \in \mathbb{R}^{M \times h}, \quad (7)$$

where each row $(\mathbf{z}_{r_{in}})_m \in \mathbb{R}^h$ is the output embedding produced by fusion expert m .

The output of the MoE projection layer is then computed in vectorized form as:

$$\mathbf{S}_{r_{in}} = \pi_{r_{in}}^\top \mathbf{Z}_{r_{in}} \in \mathbb{R}^h. \quad (8)$$

5.2. Topology-Guided Sparse Graph Mask

As electricity markets are physically coupled through cross-border transmission lines, this motivates a topology-aware modeling prior: input regions that are directly connected to the target region r_{out} typically exert a stronger impact than regions that are topologically distant. Incorporating features

from distant regions can introduce irrelevant or noisy signals, harming the generalization of the model. To explicitly encode this structure, we design a topology-guided graph mask to construct a sparse, output-region-specific connectivity pattern for aggregating regional representations.

Graph Distance. We produce *graph distance* by performing a breadth-first search (BFS) traversal on the cross-border grid topology, detailed in Appendix, Table 9. For a given output region $r_{\text{out}} \in \mathcal{R}$, we define the graph distance $d(r_{\text{in}}, r_{\text{out}})$ as the minimal number of transmission hops from each input region r_{in} to the output region r_{out} , based on direct or indirect physical connectivity:

$$d(r_{\text{in}}, r_{\text{out}}) = \begin{cases} 0 & \text{if } r_{\text{in}} = r_{\text{out}}, \\ 1 & \text{if } r_{\text{in}} \sim r_{\text{out}}, \\ 1 + \min_{r' \sim r_{\text{in}}} d(r', r_{\text{out}}) & \text{otherwise,} \end{cases} \quad (9)$$

where $r_{\text{in}} \sim r_{\text{out}}$ denotes that two regions are directly connected by a transmission line. For example, let $r_{\text{out}} = \text{AT}$. Then $d(\text{AT}, \text{AT}) = 0$. The region HU is directly connected to AT, thus $d(\text{HU}, \text{AT}) = 1$. SK is indirectly connected to AT via HU, yielding $d(\text{SK}, \text{AT}) = 2$.

Sparse Graph. If a distant region experiences an exogenous event (e.g., a surge in solar generation), its impact will first affect its neighborhood and then propagate gradually along the topology before reaching the neighborhood of the target region. Being said, to model the target region accurately, we should prioritize input features from its closer neighbors. Motivated by this propagation mechanism and the observation that distant features may be noisy, we construct a sparse graph mask to restrict information flow to a bounded neighborhood of each target region.

Specifically, for each target region $r_{\text{out}} \in \mathcal{R}$, we compute the graph distance $d(r_{\text{in}}, r_{\text{out}})$ for all input regions $r_{\text{in}} \in \mathcal{R}$ using Eq. (9) and define the output-specific mask:

$$\mathbf{m}_{r_{\text{out}}} = \begin{bmatrix} \mathbb{I}(d(\text{AT}, r_{\text{out}}) \leq \delta) \\ \mathbb{I}(d(\text{BE}, r_{\text{out}}) \leq \delta) \\ \vdots \\ \mathbb{I}(d(\text{SK}, r_{\text{out}}) \leq \delta) \end{bmatrix} \in \{0, 1\}^{|\mathcal{R}| \times 1}, \quad (10)$$

where $\mathbb{I}(\cdot)$ is the indicator function, and $\delta \in \mathbb{N}$ is the graph degree cutoff controlling the maximum neighborhood radius retained for r_{out} . By controlling δ , we can perform case studies for each target region to understand how far along the grid topology neighboring information remains beneficial.

As an example, let $r_{\text{out}} = \text{AT}$ and $\delta = 0$. Then, only AT is assigned a mask value of 1, the rest input regions are assigned 0, meaning that no information from any neighbors is used. If $\delta = 1$, then only regions directly connected to

AT are assigned a mask value of 1 (e.g., HU and SI), while all other regions with $d(r_{\text{in}}, \text{AT}) > 1$ are assigned 0.

Regional Aggregation. The regional embeddings $\{\mathbf{S}_{r_{\text{in}}}\}_{r_{\text{in}} \in \mathcal{R}}$ from Eq. (8) are stacked to form the spatial representation:

$$\mathbf{S} = \text{Stack}(\{\mathbf{S}_{r_{\text{in}}}\}_{r_{\text{in}} \in \mathcal{R}}) \in \mathbb{R}^{|\mathcal{R}| \times h}. \quad (11)$$

The topology-guided sparsity is injected into $\mathbf{S}$ by computing a sparsity-constrained average representation over the masked neighborhood of each target region r_{out} :

$$\mathbf{U}_{r_{\text{out}}} = \frac{\mathbf{m}_{r_{\text{out}}}^\top \mathbf{S}}{\mathbf{m}_{r_{\text{out}}}^\top \mathbf{1}}, \quad (12)$$

where $\mathbf{1} \in \mathbb{R}^{|\mathcal{R}| \times 1}$ is a vector of ones.

This operation acts as spatial regularization by restricting aggregation to a physically plausible neighborhood.

5.3. Hierarchical Head

We design a multi-region, multi-timestep, and multi-quantile head. To prevent quantile crossing issue¹, we adopt a hierarchical quantile head (Yu et al., 2026b). Specifically, the median quantile ($\tau_m = 0.5$) price trajectory, which represents the full set of timesteps $\mathcal{T}$, is predicted from $\mathbf{U}_{r_{\text{out}}}$ via a dense layer $\mathcal{F}_{\tau_m}(\cdot)$:

$$\hat{\mathbf{y}}_{r_{\text{out}}, \tau_m} = \mathcal{F}_{\tau_m}(\mathbf{U}_{r_{\text{out}}}) \in \mathbb{R}^{\mathcal{T}}. \quad (13)$$

To produce the upper quantile forecast ($\tau_u > 0.50$), a residual price trajectory $\hat{\mathbf{r}}_{r_{\text{out}}, \tau_u}$ is generated from $\mathbf{U}_{r_{\text{out}}}$:

$$\hat{\mathbf{r}}_{r_{\text{out}}, \tau_u} = \mathcal{F}_{\tau_u}(\mathbf{U}_{r_{\text{out}}}) \in \mathbb{R}^{\mathcal{T}}, \quad (14)$$

where a non-negative function $g(\cdot)$, such as absolute-value function, is applied to the price residual. The final upper quantile forecast is obtained by adding this non-negative residual to the median:

$$\hat{\mathbf{y}}_{r_{\text{out}}, \tau_u} = \hat{\mathbf{y}}_{r_{\text{out}}, \tau_m} + g(\hat{\mathbf{r}}_{r_{\text{out}}, \tau_u}). \quad (15)$$

For the lower quantile ($\tau_l < 0.50$), we compute a residual trajectory similarly:

$$\hat{\mathbf{r}}_{r_{\text{out}}, \tau_l} = \mathcal{F}_{\tau_l}(\mathbf{U}_{r_{\text{out}}}) \in \mathbb{R}^{\mathcal{T}}, \quad (16)$$

and subtract it from the median to obtain the lower quantile prediction:

$$\hat{\mathbf{y}}_{r_{\text{out}}, \tau_l} = \hat{\mathbf{y}}_{r_{\text{out}}, \tau_m} - g(\hat{\mathbf{r}}_{r_{\text{out}}, \tau_l}). \quad (17)$$

This hierarchical design guarantees that the upper quantile prediction is greater than or equal to the lower one at each time step, overcoming quantile crossing.

¹Quantile crossing refers to the phenomenon where upper quantile predictions (e.g., 90%) fall below lower quantiles (e.g., 10%), violating the monotonicity of the quantile function.

Table 1. Zero-shot inference. The metrics are shown as mean $\pm$ standard deviation over 5 independent runs. The gray text color in the table indicates zero standard deviation. The best result is marked in **bold** and the second best is underlined. The units of AQL, AIW, RMSE, and MAE are expressed in €/MWh, while AQCR and AQCE are in %. The symbols S, M, and L denote the small, base, and large size of Moirai. * indicates that TimesFM supports only a fixed set of quantiles ($\tau \in \mathcal{Q} = \{0.10, 0.20, 0.30, 0.40, 0.50, 0.60, 0.70, 0.80, 0.90\}$); therefore, PriceFM* is evaluated using the same quantile set against TimesFM* for a fair comparison.

Model	AQL $\downarrow$	AQCR $\downarrow$	AQCE $\downarrow$	AIW $\downarrow$	RMSE $\downarrow$	MAE $\downarrow$	R ² $\uparrow$
CHRONOS	11.14 $\pm$ 0.00	0.00 $\pm$ 0.00	9.20 $\pm$ 0.00	33.79 $\pm$ 0.00	42.19 $\pm$ 0.00	25.98 $\pm$ 0.00	0.12 $\pm$ 0.00
CHRONOS [2.0]	<u>8.03$\pm$0.00</u>	0.00 $\pm$ 0.00	<u>7.59$\pm$0.00</u>	<u>26.01$\pm$0.00</u>	<u>30.93$\pm$0.00</u>	<u>19.44$\pm$0.00</u>	<u>0.48$\pm$0.00</u>
MOIRAI [S]	11.24 $\pm$ 0.00	0.00 $\pm$ 0.00	7.67 $\pm$ 0.00	38.00 $\pm$ 0.00	43.66 $\pm$ 0.00	27.22 $\pm$ 0.00	0.07 $\pm$ 0.00
MOIRAI [M]	12.07 $\pm$ 0.00	0.00 $\pm$ 0.00	<u>7.03$\pm$0.00</u>	42.00 $\pm$ 0.00	47.94 $\pm$ 0.00	30.47 $\pm$ 0.00	-0.12 $\pm$ 0.00
MOIRAI [L]	11.94 $\pm$ 0.00	0.00 $\pm$ 0.00	9.14 $\pm$ 0.00	37.15 $\pm$ 0.00	46.66 $\pm$ 0.00	29.13 $\pm$ 0.00	-0.07 $\pm$ 0.00
TIMEMOE	-	-	-	-	40.83 $\pm$ 0.00	25.54 $\pm$ 0.00	0.16 $\pm$ 0.00
TIMESFM [2.0]*	10.50 $\pm$ 0.00	0.00 $\pm$ 0.00	<u>5.43$\pm$0.00</u>	40.83 $\pm$ 0.00	41.91 $\pm$ 0.00	26.01 $\pm$ 0.00	0.15 $\pm$ 0.00
TIMESFM [2.5]*	<u>7.97$\pm$0.00</u>	0.00 $\pm$ 0.00	7.62 $\pm$ 0.00	<u>25.98$\pm$0.00</u>	<u>30.83$\pm$0.00</u>	<u>19.48$\pm$0.00</u>	<u>0.48$\pm$0.00</u>
PRICEFM	6.85$\pm$0.13	0.00$\pm$0.00	5.30$\pm$0.34	25.69$\pm$0.87	26.13$\pm$0.72	16.83$\pm$0.50	0.55$\pm$0.01
PRICEFM*	6.91$\pm$0.10	0.00$\pm$0.00	5.39$\pm$0.30	25.88$\pm$0.93	26.24$\pm$0.79	16.90$\pm$0.61	0.55$\pm$0.01

5.4. Loss

We use the *Average Quantile Loss (AQL)* as the training objective for multi-region, multi-timestep, and multi-quantile probabilistic forecasting. Let $y_{i,r_{\text{out}},t}$ denote the ground-truth price for the i -th training sample, output region r_{out} , and timestep t , and let $\hat{y}_{i,r_{\text{out}},t,\tau}$ be the corresponding predicted quantile. The AQL is computed as:

$$\text{AQL} = \frac{1}{N|\mathcal{R}||\mathcal{T}||\mathcal{Q}|} \sum_{i=1}^N \sum_{r_{\text{out}} \in \mathcal{R}} \sum_{t=1}^T \sum_{\tau \in \mathcal{Q}} L_{\tau}(y_{i,r_{\text{out}},t}, \hat{y}_{i,r_{\text{out}},t,\tau}), \quad (18)$$

where N is the number of samples, and the quantile loss L_{τ} is defined as:

$$L_{\tau}(y, \hat{y}_{\tau}) = \begin{cases} \tau \cdot (y - \hat{y}_{\tau}), & \text{if } y \geq \hat{y}_{\tau}, \\ (1 - \tau) \cdot (\hat{y}_{\tau} - y), & \text{otherwise,} \end{cases} \quad (19)$$

where y and $\hat{y}$ are the true and predicted values, respectively.

6. Experiments

6.1. Experimental Settings

Rolling Evaluation. We adopt a 3-fold rolling evaluation. In fold 1, the data span from 1. Jan 2022 to 1. Sep 2024 for training, 1. Sep 2024 to 1. Jan 2025 for validation, and 1. Jan 2025 to 1. May 2025 for testing. Each subsequent fold advances by 4 months, ending at 1. Jan 2026, so that the testing windows jointly cover one full year.

Evaluation Metrics. To evaluate the performance of pointwise forecasting, we use Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Coefficient of Determination (R²). For probabilistic forecasting, we utilize AQL, Average Quantile Crossing Rate (AQCR), Average

Quantile Coverage Error (AQCE), and Average Interval Width (AIW). The Diebold-Mariano (DM) Test is used to determine if a significant difference of two models exists. All metrics are detailed in Appendix I.

Baseline Models. We assess the performance of PriceFM under both *zero-shot* and *full-shot* settings. In the zero-shot evaluation, a leave-one-region-out strategy is applied, i.e., PriceFM is not trained on the target region. We include several pretrained time-series foundation models: **Chronos** (original and 2.0), **Moirai** (small, base, and large), **TimesFM** (2.0 and 2.5), and **TimeMoE**. In the full-shot evaluation, all models are trained from scratch. We compare against 3 **Naïve** baselines (Appendix E), **FEDFormer**, **iTransformer**, **PatchTST**, **TimesNet**, **TimeXer**, **GCN**, **GAT**, **GraphSAGE**, **GraphDiffusion**, and **GraphARMA**. The adjacency matrix used by the graph baselines is described in Appendix G.

6.2. Zero-Shot Performance.

Table 1 shows that PriceFM delivers the best zero-shot performance, achieving 14.7% lower AQL (6.85 vs. 8.03) and 15.5% lower RMSE (26.13 vs. 30.93) than the second-best Chronos 2.0. Compared with TimesFM*, PriceFM* improves AQL by 13.3% (6.91 vs. 7.97) and reduces RMSE by 14.9% (26.24 vs. 30.83). Moreover, the consistently lower AQCE together with narrower AIW indicates better calibration without sacrificing sharpness. These gains are corroborated by probabilistic Diebold-Mariano tests, with all p -values < 0.05 and negative DM values.

Table 2. Full-shot inference. The gray text color in the table indicates that the naïve baselines have zero standard deviation.

Model	AQL ↓	AQCR ↓	AQCE ↓	AIW ↓	RMSE ↓	MAE ↓	R ² ↑
NAÏVE ¹	15.29±0.00	0.00±0.00	11.34±0.00	108.02±0.00	34.68±0.00	22.06±0.00	0.33±0.00
NAÏVE ²	15.35±0.00	0.00±0.00	11.80±0.00	103.56±0.00	34.31±0.00	23.31±0.00	0.35±0.00
NAÏVE ³	15.46±0.00	0.00±0.00	12.26±0.00	101.26±0.00	32.61±0.00	22.64±0.00	0.39±0.00
FEDFORMER	8.22±0.43	15.33±0.73	8.34±0.42	25.20±0.57	31.75±1.02	20.15±0.75	0.38±0.01
PATCHTST	8.06±0.50	18.21±0.88	7.92±0.50	24.99±0.60	31.59±0.94	20.20±0.66	0.39±0.01
ITRANSFORMER	8.24±0.61	13.96±0.59	8.11±0.38	24.65±0.53	32.11±1.04	21.03±0.77	0.38±0.01
TIMESNET	7.98±0.37	13.42±0.67	8.03±0.44	23.99±0.86	30.94±0.96	19.48±0.51	0.40±0.00
TIMEXER	8.30±0.54	14.77±0.95	9.02±0.45	25.23±0.61	31.88±1.14	21.94±0.80	0.39±0.01
GCN	6.61±0.14	6.88±0.18	7.91±0.25	23.76±0.44	25.97±0.74	16.81±0.50	0.53±0.02
GAT	7.13±0.30	10.33±0.42	8.44±0.30	24.96±0.75	26.11±0.69	17.00±0.47	0.51±0.02
GRAPHSAGE	6.78±0.18	6.01±0.06	7.44±0.21	24.18±0.47	26.03±0.99	17.56±0.60	0.53±0.01
GRAPHDIFFUSION	6.69±0.20	5.72±0.16	7.94±0.43	23.50±0.66	25.93±0.79	16.44±0.51	0.54±0.01
GRAPHARMA	6.72±0.16	6.03±0.22	8.00±0.39	23.55±0.54	25.84±1.03	16.56±0.43	0.55±0.02
PRICEFM	5.80±0.09	0.00±0.00	5.25±0.27	21.27±0.40	22.39±0.38	14.28±0.22	0.60±0.01

6.3. Full-Shot Performance.

Table 2 shows that PriceFM achieves the strongest full-shot performance across all metrics. Compared to the three naïve baselines, PriceFM reduces AQL by over 60%. The time-series models remain clearly behind, which is expected because they lack explicit spatial inductive bias; concatenating multi-region inputs along the feature dimension can easily introduce spurious correlations and overfitting. Against graph baselines, PriceFM improves AQL by 12.3% over the best-performing GNN, i.e., GCN, while also yielding better calibration, as evidenced by 29.4% lower AQCE and 9.5% sharper interval width. We attribute these gains to our sparse graph masking mechanism that explicitly controls information flow, mitigating the tendency of purely data-driven GNNs to overfit by propagating noisy signals from distant or weakly related regions.

7. Ablation Study

7.0.1. SPATIOTEMPORAL CONFIGURATIONS

- **Graph Degree Cutoff:** Spatially, we evaluate $\delta \in \{0, 1, 2, 3, \dots, 10\}$, ranging from strong constraint to weak constraint. In total, 1,254 trials are conducted to determine the optimal cutoff value for each output region individually.
- **Backward-Looking Window Size:** Temporally, we compare $L \in \{96, 288, 672\}$, corresponding to 1 day, 3 days, and 1 week. For each window size, all other hyperparameters are re-optimized.

Spatially, Figure 5 illustrates the testing loss and the distribution of optimal graph cutoff values. Temporally, the results in Table 3 indicate that the optimal backward-looking

window size is 96, potentially because information from the distant past becomes outdated.

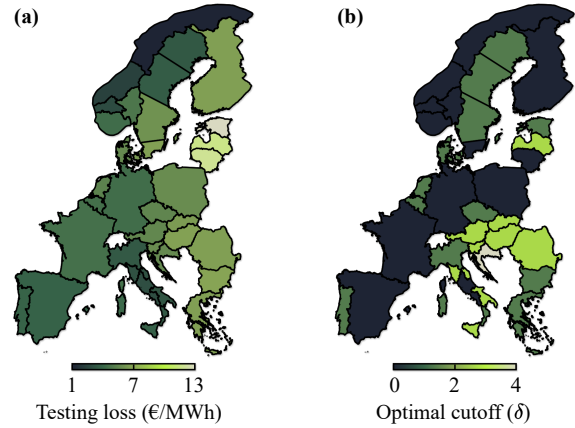


Figure 5. Spatial distribution of testing loss and graph cutoff values. (a) Average quantile loss per region on the testing set. Western and northern European regions exhibit lower losses. (b) Optimal value of graph degree cutoff per region. Notably, regions such as Germany, France, and Norway have a value of 0, indicating optimal performance by excluding neighboring features.

7.0.2. MOE PROJECTION LAYER

- **Number of Experts:** We evaluate $M \in \{1, 4, 8\}$ to study how many experts are needed to represent features from different regions under a shared projection.
- **Concatenation:** We replace the residual addition from Eq. (3) by concatenation:

$$\mathbf{z}_{r_{in}} = \text{Swish}(\text{Concat}(\hat{\mathbf{z}}_{r_{in}}^{\text{price}}, \hat{\mathbf{z}}_{r_{in}}^{\text{exo}})) \in \mathbb{R}^{2h}. \quad (20)$$

Table 3. Ablation studies of different module choices. The symbol † marks the method used in PriceFM.

Model	AQL $\downarrow$	AQCR $\downarrow$	AQCE $\downarrow$	AIW $\downarrow$	RMSE $\downarrow$	MAE $\downarrow$	R ² $\uparrow$
$L = 96$ [1 DAY] †	5.80$\pm$0.09	0.00$\pm$0.00	5.25$\pm$0.27	21.27$\pm$0.40	22.39$\pm$0.38	14.28$\pm$0.22	0.60$\pm$0.01
$L = 288$ [3 DAYS]	5.86 $\pm$ 0.11	0.00$\pm$0.00	6.34 $\pm$ 0.33	22.15 $\pm$ 0.30	22.51 $\pm$ 0.56	14.30 $\pm$ 0.19	0.58 $\pm$ 0.00
$L = 672$ [1 WEEK]	5.96 $\pm$ 0.15	0.00$\pm$0.00	7.34 $\pm$ 0.39	22.65 $\pm$ 0.41	23.83 $\pm$ 0.64	15.01 $\pm$ 0.23	0.57 $\pm$ 0.01
$M = 1$	6.15 $\pm$ 0.10	0.00$\pm$0.00	6.12 $\pm$ 0.29	22.33 $\pm$ 0.45	22.56 $\pm$ 0.40	14.47 $\pm$ 0.22	0.58 $\pm$ 0.00
$M = 4$ †	5.80$\pm$0.09	0.00$\pm$0.00	5.25 $\pm$ 0.27	21.27 $\pm$ 0.40	22.39$\pm$0.38	14.28$\pm$0.22	0.60 $\pm$ 0.01
$M = 8$	5.81 $\pm$ 0.09	0.00$\pm$0.00	5.23$\pm$0.25	21.26$\pm$0.36	22.42 $\pm$ 0.41	14.30 $\pm$ 0.24	0.60$\pm$0.00
RES. ADD †	5.80 $\pm$ 0.09	0.00$\pm$0.00	5.25 $\pm$ 0.27	21.27$\pm$0.40	22.39$\pm$0.38	14.28$\pm$0.22	0.60$\pm$0.01
CONCAT.	6.11 $\pm$ 0.14	0.00$\pm$0.00	7.34 $\pm$ 0.36	21.90 $\pm$ 0.52	23.03 $\pm$ 0.44	14.80 $\pm$ 0.37	0.59 $\pm$ 0.01
CROSS-ATTN	5.79$\pm$0.09	0.00$\pm$0.00	5.24$\pm$0.29	21.30 $\pm$ 0.42	22.41 $\pm$ 0.38	14.33 $\pm$ 0.31	0.60 $\pm$ 0.01
SPARSE GRAPH †	5.80$\pm$0.09	0.00$\pm$0.00	5.25$\pm$0.27	21.27$\pm$0.40	22.39$\pm$0.38	14.28$\pm$0.22	0.60$\pm$0.01
RANDOM	7.23 $\pm$ 0.22	0.00$\pm$0.00	8.44 $\pm$ 0.31	24.99 $\pm$ 0.70	26.13 $\pm$ 0.67	17.05 $\pm$ 0.48	0.51 $\pm$ 0.02
NO MASK	6.65 $\pm$ 0.16	0.00$\pm$0.00	7.93 $\pm$ 0.20	23.84 $\pm$ 0.52	25.82 $\pm$ 0.66	16.41 $\pm$ 0.43	0.55 $\pm$ 0.01
ABSOLUTE †	5.80$\pm$0.09	0.00$\pm$0.00	5.25 $\pm$ 0.27	21.27$\pm$0.40	22.39 $\pm$ 0.38	14.28 $\pm$ 0.22	0.60 $\pm$ 0.01
ReLU	5.80 $\pm$ 0.11	0.00$\pm$0.00	5.24$\pm$0.28	21.30 $\pm$ 0.37	22.40 $\pm$ 0.41	14.29 $\pm$ 0.18	0.60$\pm$0.00
STANDARD	5.81 $\pm$ 0.08	5.04 $\pm$ 0.12	5.26 $\pm$ 0.25	21.27 $\pm$ 0.43	22.39$\pm$0.37	14.26$\pm$0.17	0.60 $\pm$ 0.01

- **Cross-Attention:** We apply multi-head attention with $\mathbf{X}_{r_{in}}^{\text{price}}$ as the query and $\mathbf{X}_{r_{in}}^{\text{exo}}$ as both key and value to produce the attention fused feature:

$$\mathbf{z}_{r_{in}} = \text{CrossAttention}(\mathbf{X}_{r_{in}}^{\text{price}}, \mathbf{X}_{r_{in}}^{\text{attn}}). \quad (21)$$

The results in Table 3 show that using a single expert yields 6.0% higher AQL than using $M = 4$. Further increasing the number of experts to $M = 8$ does not further improve the loss. This indicates that $M = 4$ is sufficient to differentiate regional patterns. Replacing the residual addition with concatenation leads to 5.3% higher AQL and switching to cross-attention yields comparable performance to residual addition, while introducing additional parameters. This suggests that the residual addition strikes a favorable balance between predictive performance and model simplicity.

7.0.3. TOPOLOGY-GUIDED SPARSE GRAPH MASK

- **Random Graph Mask:** We replace Eq. (10) with a randomly sampled vector, where each decay weight is drawn independently from a uniform distribution over $[0, 1]$, thereby removing the spatial graph prior:

$$\mathbf{m}_{r_{out}} \sim \mathcal{U}(0, 1)^{|\mathcal{R}| \times 1}. \quad (22)$$

- **No Graph Mask:** We remove the decay mask, which simplifies Eq. (12) to a uniform average over input regions:

$$\mathbf{U}_{r_{out}} = \frac{\mathbf{1}^\top \mathbf{S}}{|\mathcal{R}|}, \quad (23)$$

The results in Table 3 demonstrate that randomizing or removing the graph decay mask, which discards the prior

graph knowledge, leads to a significant drop in all metrics. We also observe that such results are on par with those of GNN baselines. We emphasize that relying on pure data-driven learning without an explicit graph-based constraint leads to a loss of the key inductive bias, thereby limiting the model’s performance.

7.0.4. HIERARCHICAL QUANTILE HEAD

- **Non-Negative Functions:** We replace the absolute-value function used in Eq. (15) and (17) with ReLU.
- **Standard Multi-Quantile Head:** The Eq. (14) and (16) are skipped, and $\mathbf{U}_{r_{out}}$ is passed directly to independent dense layers to produce quantile trajectories.

The results in Table 3 reveal that replacing the absolute-value function with ReLU does not result in a noticeable change in overall performance, suggesting that the choice of non-negative function is flexible. Moreover, while the hierarchical quantile head achieves comparable loss to the standard multi-quantile head, the latter exhibits a mean AQCR of 5.04%, indicating that the hierarchical design mitigates quantile crossing without harming performance.

8. Conclusion

In this paper, we introduced a comprehensive, large, and up-to-date dataset, which will benefit both the research community and the energy industry. Furthermore, we proposed PriceFM, a foundation model pretrained on this diverse dataset, showing strong generalizability in both zero-shot and full-shot settings. Extensive experiments and ablation studies highlight the importance of spatial context and individual contribution of design choices. By enabling more

accurate and comprehensive probabilistic electricity price forecasting, our work has the potential to support better decision-making in energy trading and grid management.

Impact Statement

The developed PRICEFM is a domain-specific foundation model pretrained on large and diverse European energy data, and is applicable for 38 European regions. By injecting transmission-topology priors, PriceFM achieves improved performance under both zero-shot and full-shot evaluation. Beyond accuracy, PriceFM is also designed for practical deployment in a continuously evolving power system. By leveraging large-scale pretraining and transferable representations, the model can be tuned as the European transmission grid and market conditions change, enabling flexible adaptation to structural shifts.

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A. Limitation

First, as the physical transmission network evolves over time (e.g., new interconnections or bidding-zone reconfigurations), model retraining or re-calibration may be required to reflect structural changes in inter-regional coupling. Second, our topology-guided sparse graph mask is constructed solely from the transmission connectivity prior. While this choice is physically grounded, alternative notions of spatial relatedness may further improve performance. For example, one could explore geographic proximity (e.g., a European map-based topology) or design alternative weighting schemes based on socio-economic or system characteristics such as population, load, renewable penetration, or power flow, potentially yielding better cross-region aggregation.

B. Code Guideline

We open-source all code for preprocessing, modeling, and analysis. The project directory is structured as follows:

```
| - Parent Folder/  
  | - Data/  
  | - Figure/  
  | - Model/  
  | - Result/  
  | - PriceFM/  
    | - data.py  
    | - model.py  
    | - evaluation.py  
    | - pipeline.py  
  | - Tutorial.ipynb  
  | - README.md
```

where the `README.md` specifies the required package version. To facilitate reproducibility and accessibility, we have streamlined the entire pipeline through extensive engineering efforts into just three simple steps:

Step 1: Create folders named `Data`, `Figure`, `Model`, and `Result`. Place the energy data `Final.csv` into `Data`.

Step 2: Run `Tutorial.ipynb` to understand the energy data, and to train, validate, and test the PriceFM. The script `model.py` contains all necessary functions and classes of PriceFM.

Step 3: After execution, you can inspect: `Model/` for saved model weights; `Result/` for evaluation metrics.

C. Hardware and Computation

The PriceFM is evaluated on both an NVIDIA A100 GPU and an Intel Core i7-1265U CPU, respectively. The NVIDIA A100 is designed for high-performance computing and deep learning workloads, offering 80 GB of high-bandwidth memory and up to 6,912 CUDA cores. In contrast, the Intel i7-1265U is a power-efficient CPU commonly found in standard laptops. The training time is approximately 2-3 minutes on the A100 GPU and 6-7 minutes on the i7 CPU. Inference time for both setups is under 10 seconds. We note that neither training nor inference time is critical for our application, as bid submissions can occur at any point before the market gate closure on a daily basis.

D. Lookup Table and Feature Availability

The country-region code lookup table and the feature availability are listed in Table 4.

E. Naïve Baselines

We include naïve baselines as reference models, where only historical prices are used as input: **Naïve**¹ uses 96 prices from the previous day; **Naïve**² uses 96 prices averaged over the past three days; **Naïve**³ uses 96 prices averaged over the past seven days. To obtain probabilistic results, we compute empirical quantiles at individual levels for each delivery hour. The seasonal naïves are commonly used to evaluate the autoregressive strength of the signal and often serve as strong baselines

Table 4. Lookup table and feature availability across European regions. ✓ indicates that the feature is available.

Country	Region Code	Price	Load	Solar	Wind
Austria	AT	✓	✓	✓	✓
Belgium	BE	✓	✓	✓	✓
Bulgaria	BG	✓	✓	✓	✓
Czech Republic	CZ	✓	✓	✓	
Germany, Luxembourg	DE-LU	✓	✓	✓	✓
Denmark	DK1	✓	✓	✓	✓
Denmark	DK2	✓	✓	✓	✓
Estonia	EE	✓	✓	✓	✓
Spain	ES	✓	✓	✓	✓
Finland	FI	✓	✓	✓	✓
France	FR	✓	✓	✓	✓
Greece	GR	✓	✓	✓	✓
Croatia	HR	✓	✓	✓	✓
Hungary	HU	✓	✓	✓	✓
Italy	IT-CALA	✓	✓	✓	✓
Italy	IT-CNOR	✓	✓	✓	✓
Italy	IT-CSUD	✓	✓	✓	✓
Italy	IT-NORD	✓	✓	✓	✓
Italy	IT-SARD	✓	✓	✓	✓
Italy	IT-SICI	✓	✓	✓	✓
Italy	IT-SUD	✓	✓	✓	✓
Lithuania	LT	✓	✓	✓	✓
Latvia	LV	✓	✓		✓
Netherlands	NL	✓	✓	✓	✓
Norway	NO1	✓	✓		✓
Norway	NO2	✓	✓		✓
Norway	NO3	✓	✓		✓
Norway	NO4	✓	✓		✓
Norway	NO5	✓	✓		
Poland	PL	✓	✓	✓	✓
Portugal	PT	✓	✓	✓	✓
Romania	RO	✓	✓	✓	✓
Sweden	SE1	✓	✓	✓	✓
Sweden	SE2	✓	✓	✓	✓
Sweden	SE3	✓	✓	✓	✓
Sweden	SE4	✓	✓	✓	✓
Slovenia	SI	✓	✓	✓	
Slovakia	SK	✓	✓	✓	

(Ziel & Weron, 2018; Lago et al., 2021).

F. Hyperparameter Optimization

All models are optimized based on validation loss, and the checkpoint with the lowest validation loss is saved. We use the Adam optimizer with a default learning rate of 1×10^{-3} . Models are trained for 20 epochs with a batch size of 128. We empirically vary the learning rate to 1×10^{-4} and 4×10^{-3} , and the batch size to 32 and 64, and observe that for batch sizes ≤ 128 , the lowest validation loss across all models can consistently be reached within 20 epochs; moreover, smaller batch sizes typically converge in fewer epochs. The search space of other hyperparameters varies by model and is summarized in Table 5, Table 6, and Table 7, respectively.

Table 5. Hyperparameter search space for PriceFM.

Model	Search Space
PriceFM	hidden_size: {24, 72, 168} n_layers: {2, 3, 4} n_experts: {1, 4, 8} graph_degree_cutoff: {0, 1, 2, 3, ..., 10}

G. Adjacency Matrix

We model the European market as a graph $G = (\mathcal{R}, \mathcal{E})$, where each node $r \in \mathcal{R}$ is a bidding zone and edges indicate direct power flow via cross-border interconnections. This spatial topology is detailed in Table 9. Let $\mathcal{N}(r)$ denote the set of directly connected neighbors of r , excluding r itself. The binary adjacency matrix $A \in \{0, 1\}^{|\mathcal{R}| \times |\mathcal{R}|}$ is defined by

$$A_{r,s} = \begin{cases} 1, & \text{if } s \in \mathcal{N}(r), \\ 0, & \text{otherwise,} \end{cases} \quad r, s \in \mathcal{R}. \quad (24)$$

For GNN layers, self-loops can be added via $\tilde{A} = A + I$.

H. Data Scaling

To normalize the data while being robust to extreme values, we employ a `RobustScaler` fitted on the training data, using the `Scikit-Learn` implementation. The fitted scaler is then used to transform validation and testing data.

I. Metrics

I.1. Average Quantile Crossing Rate (AQCR)

AQCR captures the proportion of symmetric quantile pairs that violate quantile monotonicity, i.e., when a lower quantile prediction exceeds its corresponding higher quantile prediction. Let $\mathcal{P}_{\text{sym}} = \{(\tau_l, \tau_u) \in \mathcal{Q} \times \mathcal{Q} \mid \tau_l < \tau_u, \tau_l + \tau_u = 1\}$ denote the set of symmetric quantile pairs. For each prediction instance (i, r, t) and each $(\tau_l, \tau_u) \in \mathcal{P}_{\text{sym}}$, we define the crossing indicator as:

$$C_{i,r,t,\tau_l,\tau_u} = \mathbb{I}(\hat{y}_{i,r,t,\tau_l} > \hat{y}_{i,r,t,\tau_u}), \quad (25)$$

where $\mathbb{I}(\cdot)$ is an indicator function that returns 1 if the condition holds and 0 otherwise.

We compute the AQCR as:

$$\text{AQCR} = \frac{1}{N|\mathcal{R}|\mathcal{T}|\mathcal{P}_{\text{sym}}|} \sum_{i=1}^N \sum_{r \in \mathcal{R}} \sum_{t=1}^{\mathcal{T}} \sum_{(\tau_l, \tau_u) \in \mathcal{P}_{\text{sym}}} C_{i,r,t,\tau_l,\tau_u}. \quad (26)$$

A lower AQCR indicates fewer quantile crossing violations and thus reflects more reliable probabilistic forecasts.

I.2. Average Quantile Coverage Error (AQCE)

AQCE measures the calibration error of predictive intervals induced by symmetric quantile pairs. Let $\mathcal{P}_{\text{sym}} = \{(\tau_l, \tau_u) \in \mathcal{Q} \times \mathcal{Q} \mid \tau_l < \tau_u, \tau_l + \tau_u = 1\}$. For each prediction instance (i, r, t) and each $(\tau_l, \tau_u) \in \mathcal{P}_{\text{sym}}$, we define the interval coverage indicator as:

$$\Gamma_{i,r,t,\tau_l,\tau_u} = \mathbb{I}(\hat{y}_{i,r,t,\tau_l} \leq y_{i,r,t} \leq \hat{y}_{i,r,t,\tau_u}), \quad (27)$$

where $\mathbb{I}(\cdot)$ is an indicator function. We compute AQCE as:

$$\text{AQCE} = \frac{1}{|\mathcal{P}_{\text{sym}}|} \sum_{(\tau_l, \tau_u) \in \mathcal{P}_{\text{sym}}} \left| \frac{1}{N|\mathcal{R}|\mathcal{T}} \sum_{i=1}^N \sum_{r \in \mathcal{R}} \sum_{t=1}^{\mathcal{T}} \Gamma_{i,r,t,\tau_l,\tau_u} - (\tau_u - \tau_l) \right|. \quad (28)$$

A lower AQCE indicates better calibrated predictive intervals.

Table 6. Hyperparameter search space for time-series models.

Model	Search Space
FEDFormer	hidden_size: {32, 128, 512} conv_hidden_size: {32, 128, 512} e_layers: {2, 3, 4} n_heads: {2, 4, 8} dropout: {0.1, 0.3, 0.5}
iTransformer	hidden_size: {32, 128, 512} e_layers: {2, 3, 4} d_ff: {512, 1024, 2048} n_heads: {2, 4, 8} dropout: {0.1, 0.3, 0.5}
PatchTST	hidden_size: {32, 128, 512} e_layers: {2, 3, 4} n_heads: {2, 4, 8} dropout: {0.1, 0.3, 0.5} patch_len: {4, 6, 12}
TimesNet	hidden_size: {32, 128, 512} conv_hidden_size: {32, 128, 512} e_layers: {2, 3, 4} dropout: {0.1, 0.3, 0.5}
TimeXer	hidden_size: {32, 128, 512} e_layers: {2, 3, 4} n_heads: {2, 4, 8} d_ff: {512, 1024, 2048} dropout: {0.1, 0.3, 0.5}

I.3. Average Interval Width (AIW)

AIW measures the sharpness of predictive intervals induced by symmetric quantile pairs. For each prediction instance (i, r, t) and each $(\tau_l, \tau_u) \in \mathcal{P}_{\text{sym}}$, we define the interval width as:

$$W_{i,r,t,\tau_l,\tau_u} = |\hat{y}_{i,r,t,\tau_u} - \hat{y}_{i,r,t,\tau_l}|. \quad (29)$$

We compute AIW as:

$$\text{AIW} = \frac{1}{N|\mathcal{R}||\mathcal{T}||\mathcal{P}_{\text{sym}}|} \sum_{i=1}^N \sum_{r \in \mathcal{R}} \sum_{t=1}^{\mathcal{T}} \sum_{(\tau_l, \tau_u) \in \mathcal{P}_{\text{sym}}} W_{i,r,t,\tau_l,\tau_u}. \quad (30)$$

A lower AIW indicates sharper (narrower) predictive intervals.

I.4. Root Mean Squared Error (RMSE)

We compute RMSE within each region, then average over all regions:

$$\text{RMSE}_r = \sqrt{\frac{1}{N\mathcal{T}} \sum_{i=1}^N \sum_{t=1}^{\mathcal{T}} (y_{i,r,t} - \hat{y}_{i,r,t,0.5})^2}, \quad (31)$$

$$\text{RMSE} = \frac{1}{|\mathcal{R}|} \sum_{r \in \mathcal{R}} \text{RMSE}_r. \quad (32)$$

Table 7. Hyperparameter search space for spatial models.

Model	Search Space
GCN	hidden_size: {32, 128, 512} layers: {2, 3, 4} dropout: {0.1, 0.3, 0.5}
GAT	hidden_size: {32, 128, 512} layers: {2, 3, 4} n_heads: {2, 4, 8} dropout: {0.1, 0.3, 0.5}
GraphSAGE	hidden_size: {32, 128, 512} layers: {2, 3, 4} aggregate: {mean, max, sum}
GraphDiff	diff_steps: {2, 4, 6} hidden_size: {32, 128, 512} layers: {2, 3, 4}
GraphARMA	hidden_size: {32, 128, 512} layers: {2, 3, 4} order: {1, 2, 4} iteration: {1, 2, 4}

Table 8. Model capability comparison.

Model	Multivariate Input	Probabilistic Output
Chronos		✓
Chronos [2.0]	✓	✓
Moirai [S]	✓	✓
Moirai [M]	✓	✓
Moirai [L]	✓	✓
TimeMoE		
TimesFM [2.0]		✓
TimesFM [2.5]	✓	✓
PriceFM	✓	✓

I.5. Mean Absolute Error (MAE)

Same procedure as RMSE, but using absolute error:

$$\text{MAE}_r = \frac{1}{N\mathcal{T}} \sum_{i=1}^N \sum_{t=1}^{\mathcal{T}} |y_{i,r,t} - \hat{y}_{i,r,t,0.5}|, \quad (33)$$

$$\text{MAE} = \frac{1}{|\mathcal{R}|} \sum_{r \in \mathcal{R}} \text{MAE}_r. \quad (34)$$

I.6. Coefficient of Determination

We compute the Coefficient of Determination (R^2) for each region and average across all regions:

$$R_r^2 = 1 - \frac{\sum_{i=1}^N \sum_{t=1}^{\mathcal{T}} (y_{i,r,t} - \hat{y}_{i,r,t,0.5})^2}{\sum_{i=1}^N \sum_{t=1}^{\mathcal{T}} (y_{i,r,t} - \bar{y}_r)^2}, \quad (35)$$

$$\bar{y}_r = \frac{1}{N\mathcal{T}} \sum_{i=1}^N \sum_{t=1}^{\mathcal{T}} y_{i,r,t}, \quad (36)$$

$$R^2 = \frac{1}{|\mathcal{R}|} \sum_{r \in \mathcal{R}} R_r^2. \quad (37)$$

I.7. Diebold & Mariano Test / Time-Series HAC

To assess whether differences in forecasting performance are statistically significant, we apply the Diebold–Mariano (DM) test using a heteroskedasticity and autocorrelation consistent (HAC) variance estimator. Let model $l \in \{1, 2\}$ denote two competing forecasters. We define the loss differential as *loss(model 1) minus loss(model 2)*, so that a positive mean differential indicates that model 2 performs better on average.

Loss differential time series. For probabilistic forecasts, we first compute the per-instance differential for each (i, r, t, τ) :

$$d_{i,r,t,\tau} = L_{\tau}\left(y_{i,r,t}, \hat{y}_{i,r,t,\tau}^{(1)}\right) - L_{\tau}\left(y_{i,r,t}, \hat{y}_{i,r,t,\tau}^{(2)}\right), \quad (38)$$

and aggregate it across regions, horizons, and quantiles to obtain a *chronological* loss differential series $\{d_i\}_{i=1}^N$:

$$d_i = \frac{1}{|\mathcal{R}| |\mathcal{T}| |\mathcal{Q}|} \sum_{r \in \mathcal{R}} \sum_{t=1}^{\mathcal{T}} \sum_{\tau \in \mathcal{Q}} d_{i,r,t,\tau}. \quad (39)$$

HAC variance and DM statistic. Let $\bar{d} = \frac{1}{N} \sum_{i=1}^N d_i$. To account for serial correlation (e.g., induced by overlapping $\mathcal{T}$ -step forecasts), we estimate the long-run variance via Newey–West with Bartlett weights:

$$\hat{\gamma}_{\ell} = \frac{1}{N} \sum_{i=\ell+1}^N (d_i - \bar{d})(d_{i-\ell} - \bar{d}), \quad \ell = 0, 1, \dots, B, \quad (40)$$

$$\hat{\sigma}_{\text{LR}}^2 = \hat{\gamma}_0 + 2 \sum_{\ell=1}^B \left(1 - \frac{\ell}{B+1}\right) \hat{\gamma}_{\ell}, \quad (41)$$

where B denotes the Newey–West bandwidth (truncation lag). In our experiments, we set $B = \mathcal{T} - 1$.

The DM statistic is:

$$\text{DM} = \frac{\bar{d}}{\sqrt{\hat{\sigma}_{\text{LR}}^2 / N}}. \quad (42)$$

Decision rule. If $p < 0.05$ and $\text{DM} > 0$, then model 2 significantly outperforms model 1. If $p < 0.05$ and $\text{DM} < 0$, then model 1 significantly outperforms model 2. Otherwise, the difference is not statistically significant.

Table 9. Direct neighbors by region and neighbor count.

Region Code	Direct Neighbors	Count
AT	CZ, DE-LU, HU, IT-NORD, SI	5
BE	DE-LU, FR, NL	3
BG	GR, RO	2
CZ	AT, DE-LU, PL, SK	4
DE-LU	AT, BE, CZ, DK1, DK2, FR, NL, NO2, PL, SE4	10
DK1	DE-LU, DK2, NL, NO2, SE3	5
DK2	DE-LU, DK1, SE4	3
EE	FI, LV	2
ES	FR, PT	2
FI	EE, NO4, SE1, SE3	4
FR	BE, DE-LU, ES, IT-NORD	4
GR	BG, IT-SUD	2
HR	HU, SI	2
HU	AT, HR, RO, SI, SK	5
IT-CALA	IT-SICI, IT-SUD	2
IT-CNOR	IT-CSUD, IT-NORD	2
IT-CSUD	IT-CNOR, IT-SARD, IT-SUD	3
IT-NORD	AT, FR, IT-CNOR, SI	4
IT-SARD	IT-CSUD	1
IT-SICI	IT-CALA	1
IT-SUD	GR, IT-CALA, IT-CSUD	3
LT	LV, PL, SE4	3
LV	EE, LT	2
NL	BE, DK1, DE-LU, NO2	4
NO1	NO2, NO3, NO5, SE3	4
NO2	DE-LU, DK1, NL, NO1, NO5	5
NO3	NO1, NO4, NO5, SE2	4
NO4	FI, NO3, SE1, SE2	4
NO5	NO1, NO2, NO3	3
PL	CZ, DE-LU, LT, SE4, SK	5
PT	ES	1
RO	BG, HU	2
SE1	FI, NO4, SE2	3
SE2	NO3, NO4, SE1, SE3	4
SE3	DK1, FI, NO1, SE2, SE4	5
SE4	DE-LU, DK2, LT, PL, SE3	5
SI	AT, HR, HU, IT-NORD	4
SK	CZ, HU, PL	3